Mathematical Summary: The Alvarez-Macovski Model

1 Introduction

The Alvarez-Macovski (AM) method is a fundamental technique in dual-energy computed tomography (DECT). It models the energy-dependent linear attenuation coefficient (LAC) of any material as a linear combination of two basis functions, effectively capturing the physical mechanisms of X-ray attenuation.

2 The Attenuation Model

The linear attenuation coefficient $\mu(x, y, E)$ at a specific spatial position (x, y) and X-ray energy E is decomposed into energy-independent basis coefficients and energy-dependent basis functions:

$$\mu(x, y, E) = a_1(x, y)f_1(E) + a_2(x, y)f_2(E) \tag{1}$$

Typically, the basis functions are chosen to represent the photoelectric effect and Compton scattering:

$$f_1(E) = E^{-3} \quad \text{(Photoelectric effect)} \tag{2}$$

$$f_2(E) = f_{KN}(E) \quad \text{(Compton scattering - Klein-Nishina)} \tag{3}$$

where $a_1(x, y)$ relates to the material's atomic number and $a_2(x, y)$ relates to its electron density.

3 Projection Measurements

For a polychromatic X-ray source with spectrum $\Phi(E)$, the measured transmitted intensity I along a ray path is given by the Beer-Lambert law integrated over the energy spectrum:

$$I = \int \Phi(E) \exp\left(-\int \mu(x, y, E)\, dl\right) dE \tag{4}$$

By defining the line integrals of the spatial basis coefficients as A_1 and A_2:

$$A_1 = \int a_1(x, y)\, dl \tag{5}$$

$$A_2 = \int a_2(x, y)\, dl \tag{6}$$

The projection measurement simplifies to:

$$L = \ln\left(\frac{I_0}{I}\right) = A_1 f_1(E) + A_2 f_2(E) \tag{7}$$

where I_0 is the incident X-ray intensity and L is the line integral of the total attenuation.

4 Dual-Energy Decomposition

By capturing projections at two distinct X-ray energies (or with two separate energy-bin spectra), a system of two equations with two unknowns is generated. This allows the mathematical separation of the sinograms into A_1 and A_2, removing beam-hardening artifacts and generating quantitative material maps.